

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}

```

Try uname, lscpu, lsblk, ps, top commands after this program

```

sriram@sriram:~$ nano q1.c
sriram@sriram:~$ gcc q1.c -o q1
sriram@sriram:~$ ./q1
Enter a Linux command: ls
Parent Process ID: 689
Child Process ID: 690
Child Process ID: 690
Parent Process ID: 689
ANU      example.txt      s2example.txt      week13      week16      week19.c
Demo     hello           s2exampleprogram.c.save week13.c    week16.c
OSLab    hello.c          shell             week14      week17
Week1    q1                  week              week14.c    week17.c
Week1.c  q1.c              week12            week15      week18.c
Week12.c s2                  week12.c          week15.c    week19
Child process completed.
sriram@sriram:~$ uname -a
Linux sriram 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Thu Jun 5 18:30:46 UTC 2025 x86_64 GNU/Linux
sriram@sriram:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         39 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                16
On-line CPU(s) list:   0-15
Vendor ID:             GenuineIntel
Model name:            13th Gen Intel(R) Core(TM) i7-13620H
CPU family:            6
Model:                186
Thread(s) per core:    2
Core(s) per socket:    8
Socket(s):             1
Stepping:              1
BogoMIPS:              5836.79
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtrr p
ge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
ht syscall nx pdpe1gb rdtscp lm constant_tsc rep-
good_nopl xtopology tsc_reliable nonstop_tsc cpuid
tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 p
cid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline
_timer aes xsave avx f16c rdrand hypervisor lahf_l

```

```

t bml1 avx2 smep bmi2 erms invpcid rdseed adx smap
clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xs
aves avx_vnni vnmi umip waitpkg gfni vaes vpclmulq
dq rdpid movdiri movdir64b fsrm md_clear serialize
flush_lld arch_capabilities

Virtualization features:
Virtualization:      VT-x
Hypervisor vendor:   Microsoft
Virtualization type: full
Caches (sum of all):
L1d:                 384 KiB (8 instances)
L1i:                 256 KiB (8 instances)
L2:                  10 MiB (8 instances)
L3:                  24 MiB (1 instance)
NUMA:
NUMA node(s):        1
NUMA node0 CPU(s):   0-15
Vulnerabilities:
Gather data sampling: Not affected
Itlb multihit:       Not affected
L1tf:                Not affected
Mds:                 Not affected
Meltdown:            Not affected
Mmio stale data:     Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed:            Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass:   Mitigation; Speculative Store Bypass disabled via
prctl
Spectre v1:          Mitigation; usercopy/swapgs barriers and __user po
inter sanitization
Spectre v2:          Mitigation; Enhanced / Automatic IBRS; IBPB condit
ional; RSB filling; PBSRB-eIBRS SW sequence; BHI B
HI_DISABLE
Srbds:               Not affected
Tsx async abort:     Not affected
sriram@sriram:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda  8:0 0 388.4M 1 disk
sdb  8:16 0 186M 1 disk
sdc  8:32 0 2G 0 disk [SWAP]

```

sdd 8:48 0 1T 0 disk /mnt/wslg/distro

```
sriram@sriram:~$ ps
  PID TTY          TIME CMD
  594 pts/0    00:00:00 bash
  732 pts/0    00:00:00 ps
```

```
sriram@sriram:~$ top
top - 15:39:31 up 10 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7788.2 total, 7221.7 free, 559.1 used, 159.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used. 7229.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24124	14176	10976	S	0.0	0.2	0:00.76	systemd
2	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3072	1792	1792	S	0.0	0.0	0:00.00	init
47	root	19	-1	50408	15872	14848	S	0.0	0.2	0:00.21	systemd-journal
81	systemd+	20	0	22524	14208	11648	S	0.0	0.2	0:00.07	systemd-resolve
91	root	20	0	35248	11776	8832	S	0.0	0.1	0:00.53	systemd-udev
110	root	20	0	2880	1664	1664	S	0.0	0.0	0:00.04	chronyd-starter
111	root	20	0	4464	2944	2816	S	0.0	0.0	0:00.00	cron
112	message+	20	0	8804	4352	4096	S	0.0	0.1	0:00.06	dbus-daemon
118	root	20	0	44076	28152	13824	S	0.0	0.4	0:00.27	networkd-dispat
124	root	20	0	18420	8832	7808	S	0.0	0.1	0:00.12	systemd-logind
129	root	20	0	1866280	13824	11520	S	0.0	0.2	0:00.15	wsl-pro-service
157	root	20	0	5528	2688	2560	S	0.0	0.0	0:00.02	agetty
201	syslog	20	0	220532	4864	3968	S	0.0	0.1	0:00.09	rsyslogd
205	root	20	0	5484	2560	2432	S	0.0	0.0	0:00.00	agetty
229	_chrony	20	0	24504	10624	6272	S	0.0	0.1	0:00.09	chrony
246	_chrony	20	0	12076	2232	1664	S	0.0	0.0	0:00.00	chrony
273	root	20	0	123440	31872	17408	S	0.0	0.4	0:00.19	unattended-upgr
403	root	20	0	8628	5376	4608	S	0.0	0.1	0:00.01	login
448	sriram	20	0	22324	13184	10752	S	0.0	0.2	0:00.16	systemd
450	sriram	20	0	22896	4048	2048	S	0.0	0.1	0:00.00	(sd-pam)
478	sriram	20	0	6128	5248	3712	S	0.0	0.1	0:00.04	bash
591	root	20	0	3088	1028	896	S	0.0	0.0	0:00.00	SessionLeader
592	root	20	0	3088	1164	1024	S	0.0	0.0	0:00.04	Relay(594)
594	sriram	20	0	6260	5376	3712	S	0.0	0.1	0:00.10	bash
744	sriram	20	0	8164	5888	3840	R	0.0	0.1	0:00.11	top

2. copying the contents from one file to another file

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
sriram@sriram:~$ > sample.txt          :~$ echo "Hello, this is my Operating Systems
practical." > sample.txt
sriram@sriram:~$                       :~$ cat sample.txt
Hello, this is my Operating Systems practical.
sriram@sriram:~$                       :~$ Hello, this is my Operating Systems practi
cal.
Hello,: command not found
sriram@sriram:~$                       :~$ nano q2.c
sriram@sriram:~$                       :~$ gcc q2.c -o q2
sriram@sriram:~$                       :~$ ./q2
File copied successfully.
sriram@sriram:~$                       :~$ ls
ANU      copy.txt      q2          shell      week14      week17
Demo     example.txt  q2.c       week      week14.c   week17.c
OSLab    hello       s2         week12    week15     week18.c
Week1    hello.c     s2example.txt week12.c  week15.c   week19
Week1.c  q1         s2exampleprogram.c.save week13    week16     week19.c
Week12.c q1.c       sample.txt week13.c  week16.c
sriram@sriram:~$                       :~$ cat copy.txt
Hello, this is my Operating Systems practical.
sriram@sriram:~$                       :~$ cat sample.txt
Hello, this is my Operating Systems practical.
sriram@sriram:~$                       :~$ strace --version
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx
32-mpers
sriram@sriram:~$                       :~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffd912c1478 /* 25 vars */)
= 0
brk(NULL)                               = 0x5c8d2dbea000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x7d399c74f000
access("/etc/ld.so.preload", R_OK)     = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20157, ...}) = 0
mmap(NULL, 20157, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7d399c74a000
```

2. copying the contents from one file to another file

```
statx(AT_FDCWD, "/usr/share/ca-certificates/ca", AT_STATX_SYNC_AS_STAT,
STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFD
IR|0755,, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/ca-certificates/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "\n * Copyright (c) 2004-2024 Mozilla Co"..,4096) = 0. 1048576
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORE
|SA_RESTORER, sa_restorer=0x77dcd70445cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE],
sa_flags=SA_RESTORER|SA_RESTORER, sa_restorer=0x77dcd7370445cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=SA_RESTORER|SA
_RESTORER, sa_restorer=0x77dcd70445cb0}, {sa_handler=0x5cf67e17f000, sa_mask=
[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x77dcd700445cb0},
8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=[RTMIN RT_1], sa_flags=SA_
RESTORER, sa_restorer=0x77dcd70445cb0}, {sa_handler=0x5cf67e15b5ec0, sa_mask=
[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x77dcd70445cb0},
8) = 0
ioctl(1, TIOGPT2, {c_iflag=ICRNL|IXON, c_oflag=NLO|CR0|TAB0|BS0|VT0|FF0|OP0
S|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXT
EN|ECHOCTL|ECHOCTL, ...}) = 0
brk(0x5cf67e17f000) = 0 0 = 0x5cf67e17f000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=ST
ATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=47,
|...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=47, ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = 47 EINVAL (Invalid argument)
splice2([4, F_SETPIPE_SZ, 1048576, 0] = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 47
splice(5, NULL, 1, NULL, 47, 0) = 47
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x77dcd74c6000, 12288) = 0
exit_group(0) = ?
```

sriram@sriram:~\$ █